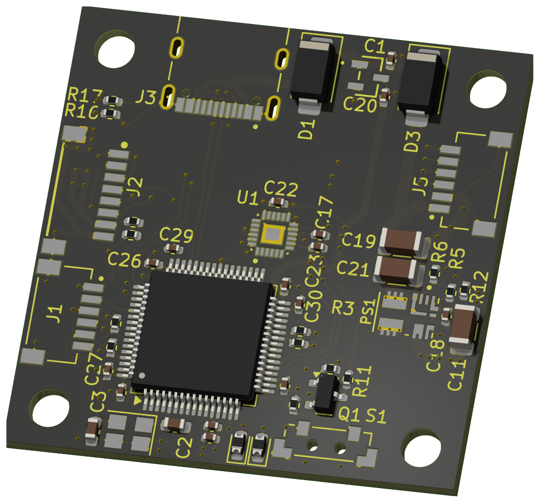
FCU3

2025-12-10

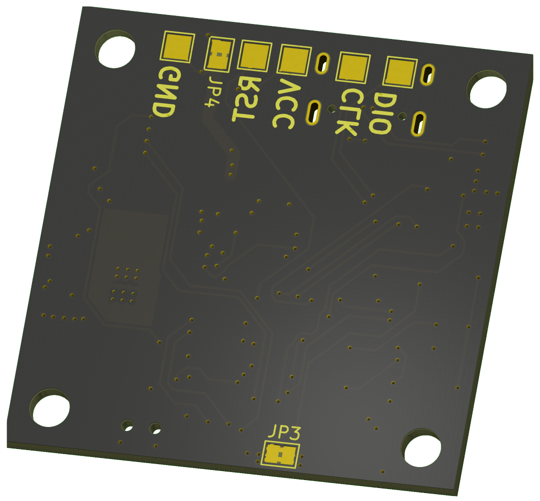
Variant: CHECKED

Rev + (Unreleased)

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 10-Dec-2025

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

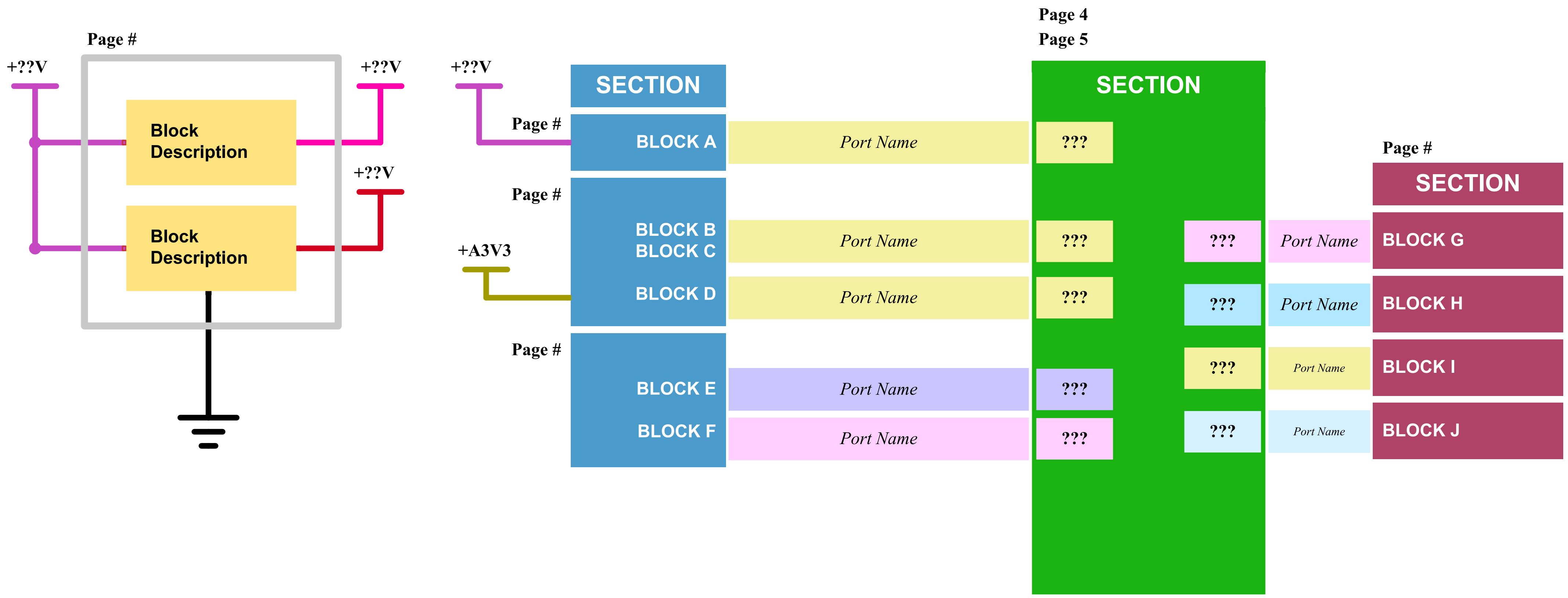
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dadb50
	Board Name: FCU3		Project Name: Flight Controller 3	
Sheet Title:	File Name: FCU3.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 8

[2] Block Diagram

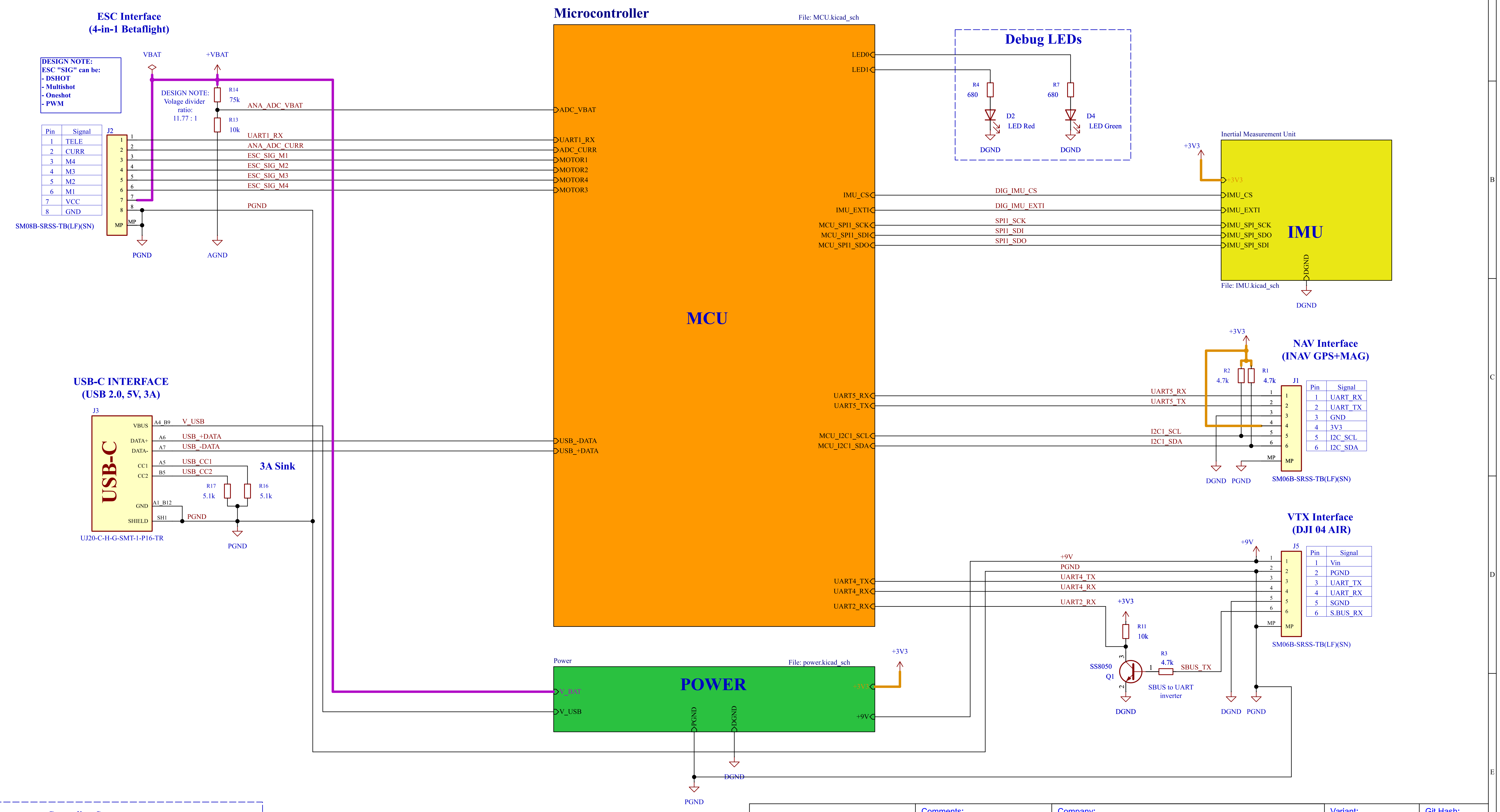


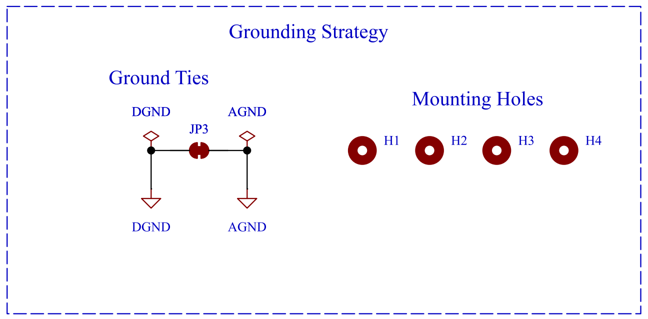
Target specifications:

Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

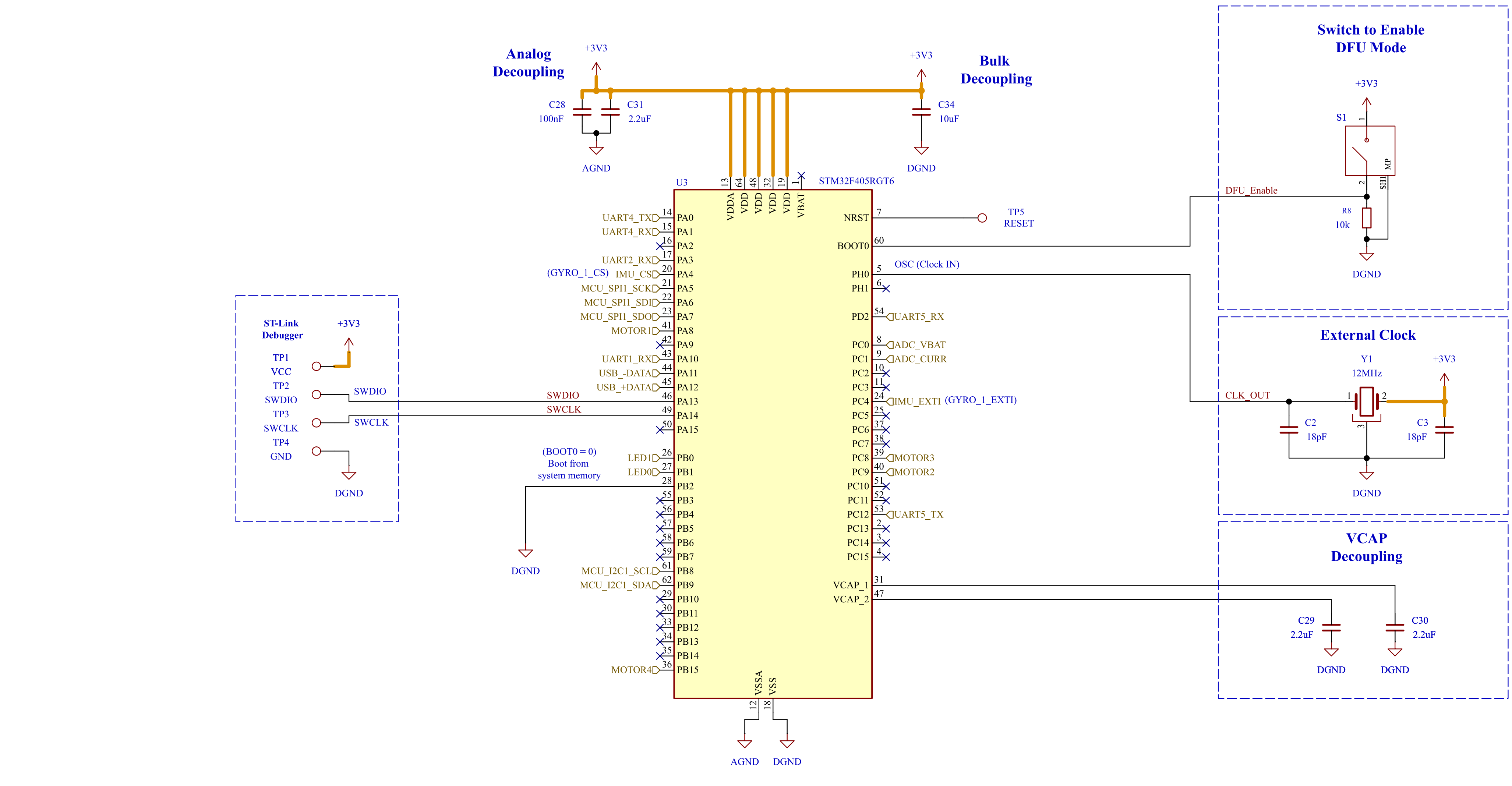
Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dadb50
	Board Name: FCU3		Project Name: Flight Controller 3	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 8

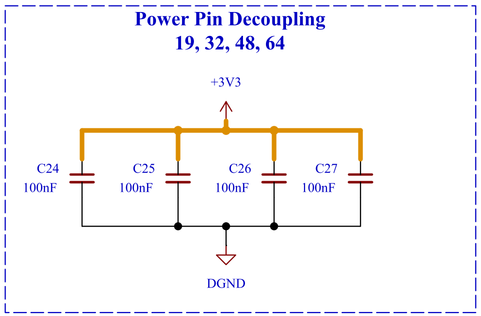
[3] Project Architecture



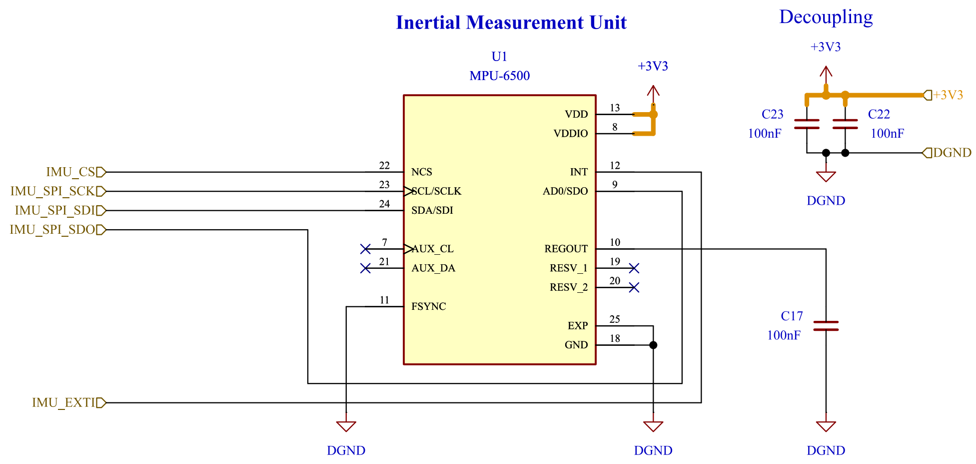


Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dadb50
	Board Name: FCU3		Project Name: Flight Controller 3	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 8



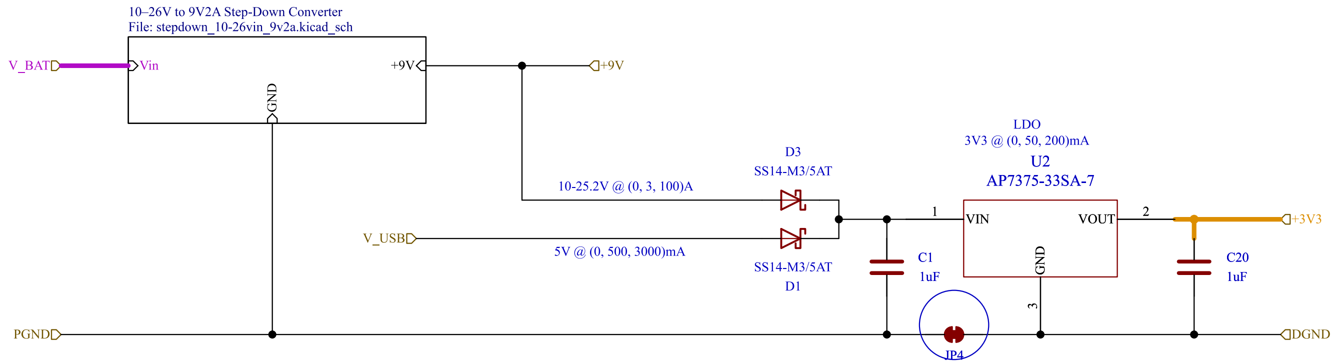


Comments:	Company:		Variant: CHECKED	Git Hash: 4dadb50
	Board Name: **FCU3**		Project Name: **Flight Controller 3**	
Sheet Title:	File Name: MCU.kicad_sch	Designer: Kaldstrand	Date:	Revision:
Sheet Path: /Project Architecture/Microcontroller/		Reviewer:	Size: **A4**	Sheet: **4** of **8**



Comments:	Company:		Variant: CHECKED	Git Hash: 4dadb50
	Board Name: **FCU3**		Project Name: **Flight Controller 3**	
Sheet Title:	File Name: IMU.kicad_sch	Designer: Kaldstrand	Date:	Revision:
Sheet Path: /Project Architecture/Inertial Measurement Unit/		Reviewer:	Size: **A4**	Sheet: **5** of **8**

To-Do:
- Add all settings used in Vbat to 9V converter.
- Add all settings used in 3.3V LDO.



DESIGN NOTE:
JP4 is the Power Ground (PGND) to
Digital Ground (DGND) connection.

Comments:

Company:

Variant: CHECKED

Git Hash: 4dadb50

Board Name: **FCU3**

Project Name: **Flight Controller 3**

Sheet Title:

File Name: power.kicad_sch

Designer: Kaldstrand

Date:

Revision:

Sheet Path: /Project Architecture/Power/

Reviewer:

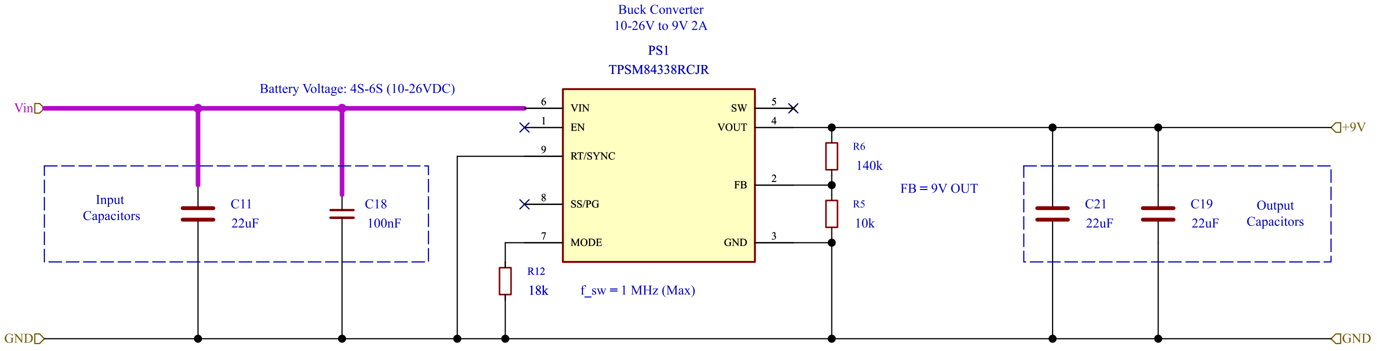
Size: **A4**

Sheet: **6** of **8**

DESIGN NOTE:
R_FBT = ((V_OUT - V_REF) / V_REF) * R_FBB
R_FBT = ((9V - 0.6V) / 0.6V) * 10k
R_FBT = 140k

V_OUT = 9V, V_REF = 0.6V, R11 = 10k, R6 = 140k

To-Do:
Add design notes
explaining DC/DC settings.



Comments:	Company:		Variant: CHECKED	Git Hash: 4dadb50
	Board Name: FCU3		Project Name: Flight Controller 3	
Sheet Title:	File Name: stepdown_10-26vin_9v2a.kicad_sch	Designer: Kaldstrand	Date:	Revision:
Sheet Path: /Project Architecture/Power/10–26V to 9V2A Step-Down Converter/		Reviewer:	Size: A4	Sheet: 8 of 8

[9] Revision History

Comments:	Company: Kaldstrand Electronics		Variant: CHECKED	Git Hash: 4dadb50
	Board Name: FCU3		Project Name: Flight Controller 3	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Kaldstrand	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 8